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Thermodynamic Assessments of the Au-Tb and Au-Lu Systems

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The thermodynamic assessments of the Au-Tb and Au-Lu binary systems were carried out by means of the CALPHAD method based on the experimental data including the thermodynamic properties and phase equilibria. The Gibbs free energies of the liquid, bcc, fcc and hcp phases were described by the substitutional solution model, while all of the intermetallic compounds (Au_6Tb, $Au_{51}Tb_{14}$, Au_3Tb, Au_2Tb, $Au_{10}Tb_7$, Au_4Tb_3, αAuTb, βAuTb, $AuTb_2$, Au_4Lu, Au_3Lu, Au_2Lu, AuLu and $AuLu_2$ phases) in these two binary systems were treated as the sublattice model. Consequently, a set of self-consistent thermodynamic parameters of each phase in the Au-Tb and Au-Lu binary systems has been obtained, and the calculated results are in good agreement with the available experimental data.

Keywords CALPHAD, phase diagram, thermodynamic assessment

1. Introduction

Gold and gold alloys are widely used in the electronic industry owing to its superior electrical and thermal properties.[1,2] The addition of the rare earth (RE) elements will further improve the mechanical properties and retain the heat conductivity and high resistance stability of gold alloys.[3,4] The calculation of phase diagrams (CALPHAD) method, which is a powerful tool to decrease cost and time during development of materials,[5] effectively provides a clear guideline for material design. As a result, it is of great essence to understand the phase equilibria in the Au-RE binary systems. In the research program, the thermodynamic database for Au-RE (RE = La, Ce, Pr, Nd, Eu, Gd, Dy, Er, Ho,…) binary systems will be developed. As a part of the Au-RE alloy system, the Au-Tb and Au-Lu binary systems were thermodynamically optimized recently in Ref 6 and 7, where the enthalpies of mixing of liquid phase[8] were not considered. On the basis of Ref 6 and 7, we reassessed the thermodynamic parameters for the phase diagrams of the Au-Tb and Au-Lu binary systems in the present work.

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2. Thermodynamic Modeling

2.1 Solution Phases

In the Au-RE (Tb or Lu) binary systems, the Gibbs free energies of the liquid, fcc, bcc, and hcp phases are described by the substitutional solution model as follows:

$$G_m^{\phi} = \sum_{i=\mathrm{Au,RE}} {}^0G_i^{\phi} x_i + RT \sum_{i=\mathrm{Au,RE}} x_i \ln x_i + \Delta^E G_m^{\phi} \qquad \text{(Eq 1)}$$

where ${}^0G_i^{\phi}$ is the reference Gibbs free energy of the pure component i in the respective reference state with ϕ phase, which is taken from the SGTE pure element database;[9] RE means the element Tb or Lu; x_i denotes the mole fraction of the component i; R is the gas constant; T is the absolute temperature; and the term $\Delta^E G_m^{\phi}$ presents the excess Gibbs free energy, which is expressed by a Redlich-Kister polynomial[10] as:

$$\Delta^E G_m^{\phi} = x_{\mathrm{Au}} x_{\mathrm{RE}} \sum_{m=0}^{n} {}^mL_{\mathrm{Au,RE}}^{\phi} (x_{\mathrm{Au}} - x_{\mathrm{RE}})^m \qquad \text{(Eq 2)}$$

$${}^mL_{Au,RE}^{\phi} = a + bT + cT \ln T \qquad \text{(Eq 3)}$$

where ${}^mL_{\mathrm{Au,RE}}^{\phi}$ is the binary interaction parameter, and the coefficients of a, b, c are constants to be optimized on the basis of available experimental data.

2.2 Stoichiometric Intermetallic Compounds

The intermetallic compounds of Au_6Tb, $Au_{51}Tb_{14}$, Au_3Tb, Au_2Tb, $Au_{10}Tb_7$, Au_4Tb_3, αAuTb, βAuTb and $AuTb_2$ phases in the Au-Tb binary system and Au_4Lu, Au_3Lu, Au_2Lu, AuLu and $AuLu_2$ phases in the Au-Lu binary system are all treated as the sublattice models.[11] The Gibbs free energy per mole of formula unit Au_pRE_q can be expressed, as follows:

$$\Delta^0 G_f^{\mathrm{Au}_p\mathrm{RE}_q} = G_{\mathrm{Au:RE}}^{\mathrm{Au}_p\mathrm{RE}_q} - p^0 G_{\mathrm{Au}}^{\mathrm{SER}} - q^0 G_{\mathrm{RE}}^{\mathrm{SER}} \qquad \text{(Eq 4)}$$

where the $\Delta^0 G_f^{Au_pRE_q}$ indicates the standard Gibbs free energy of formation of the stoichiometric compound from pure elements, which is described as:

$$\Delta^0 G_f^{Au_pRE_q} = a' + b'T + c'T\ln T + d'T^2 + e'T^3 \qquad \text{(Eq 5)}$$

where the parameters of a', b', c', d' and e' are evaluated in the present work.

3. Thermodynamic Assessments

The optimization of the thermodynamic parameters was carried out by using the PARROT program in the Thermo-Calc software,[12] which can handle various kinds of experimental data. The experimental data of the phase diagram and thermodynamic properties were used as input to the program. Each piece of selected information was given a certain weight based on the importance of data, and changed by trial and error during the assessment, until most of the selected experimental information was reproduced within the expected uncertainty limits.

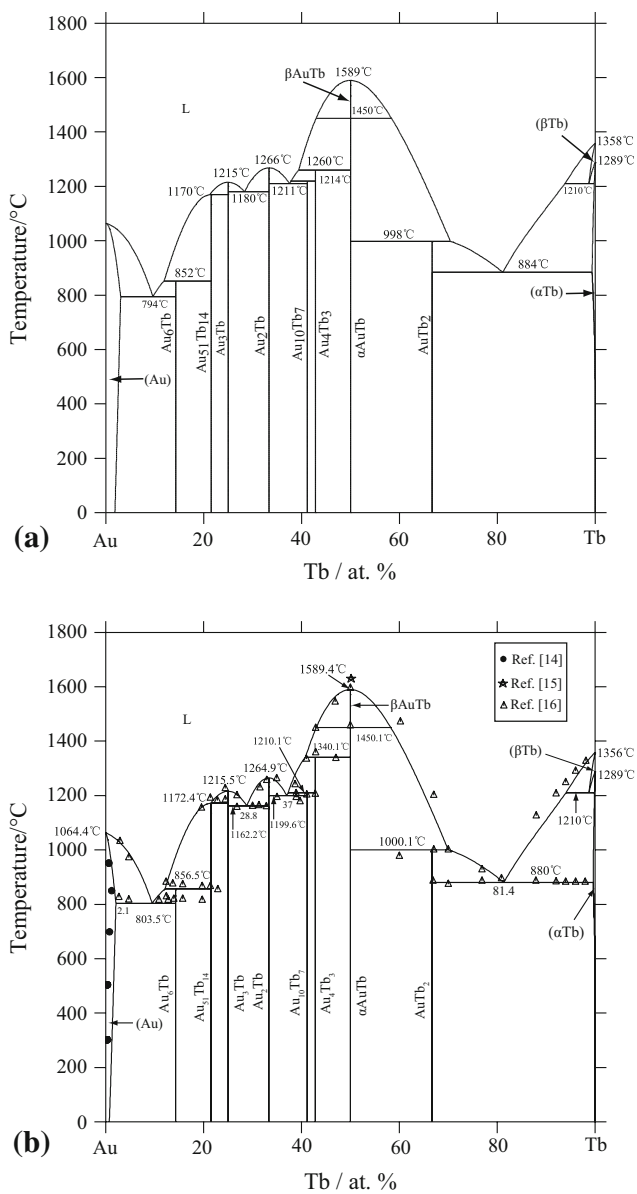

Fig. 1 Calculated phase diagram of the Au-Tb system: (a) from Ref 6; (b) in this work compared with the experimental data[14-16]

3.1 The Au-Tb System

3.1.1 Literature Information. The Au-Tb system consists of three solid solution phases (fcc (Au), bcc (βTb) and hcp (αTb) phases) and nine intermetallic compounds

Table 1 Thermodynamic parameters in the Au-Tb binary system assessed in the present work

Parameters in each phase, J/mol
Liquid phase, format (Au, Tb)
${}^0L^{Liq}_{Au,Tb}$ = - 290,500 + 5.71T
${}^1L^{Liq}_{Au,Tb}$ = - 81,100 - 92.09T + 9.0$TLn(T)$
${}^2L^{Liq}_{Au,Tb}$ = 44,080 - 7.85T
${}^3L^{Liq}_{Au,Tb}$ = 38,687 + 13.5T
Fcc phase, format (Au, Tb)
${}^1L^{fcc}_{Au,Tb}$ = $-292,672 - 3T$
Bcc phase, format (Au, Tb)
${}^1L^{bcc}_{Au,Tb}$ = 170,808
Hcp phase, format (Au, Tb)
${}^1L^{hcp}_{Au,Tb}$ = 130,000
Au_6Tb phase, format $(Au)_{0.857}$ $(Tb)_{0.143}$
${}^0G^{Au6Tb}_{Au:Tb} = 0.857 \times {}^0G^{fcc}_{Au} + 0.143 \times {}^0G^{hcp}_{Tb} - 40710 - 1.052T$
$Au_{51}Tb_{14}$ phase, format $(Au)_{0.785}$ $(Tb)_{0.215}$
${}^0G^{Au_{51}Tb_{14}}_{Au:Tb} = 0.785 \times {}^0G^{fcc}_{Au} + 0.215 \times {}^0G^{hcp}_{Tb} - 58608 - 2.578T$
Au_3Tb phase, format $(Au)_{0.75}$ $(Tb)_{0.25}$
${}^0G^{Au_3Tb}_{Au:Tb} = 0.75 \times {}^0G^{fcc}_{Au} + 0.25 \times {}^0G^{hcp}_{Tb} - 67290 - 1.16T$
Au_2Tb phase, format $(Au)_{0.667}(Tb)_{0.333}$
${}^0G^{Au_2Tb}_{Au:Tb} = 0.667 \times {}^0G^{fcc}_{Au} + 0.333 \times {}^0G^{hcp}_{Tb} - 80881 + 1.159T + 1.39TLn(T) - 8.6 \times 10^{-3}T^2 + 1.176 \times 10^{-6}T^3$
$Au_{10}Tb_7$ phase, format $(Au)_{0.588}$ $(Tb)_{0.412}$
${}^0G^{Au_{10}Tb_7}_{Au:Tb} = 0.588 \times {}^0G^{fcc}_{Au} + 0.412 \times {}^0G^{hcp}_{Tb} - 84976 + 1.7T + 1.384TLn(T) - 8.58 \times 10^{-3}T^2 + 1.176 \times 10^{-6}T^3$
Au_4Tb_3 phase, format $(Au)_{0.571}$ $(Tb)_{0.429}$
${}^0G^{Au_4Tb_3}_{Au:Tb} = 0.571 \times {}^0G^{fcc}_{Au} + 0.429 \times {}^0G^{hcp}_{Tb} - 85855 + 1.897T + 1.39TLn(T) - 8.64 \times 10^{-3}T^2 + 1.172 \times 10^{-6}T^3$
βAuTb phase, format $(Au)_{0.5}(Tb)_{0.5}$
${}^0G^{\beta AuTb}_{Au:Tb} = 0.5 \times {}^0G^{fcc}_{Au} + 0.5 \times {}^0G^{hcp}_{Tb} - 84520 + 0.12758T$
αAuTb phase, format $(Au)_{0.5}(Tb)_{0.5}$
${}^0G^{\alpha AuTb}_{Au:Tb} = 0.5 \times {}^0G^{fcc}_{Au} + 0.5 \times {}^0G^{hcp}_{Tb} - 84590 + 0.1682T$
$AuTb_2$ phase, format $(Au)_{0.333}$ $(Tb)_{0.667}$
${}^0G^{AuTb_2}_{Au:Tb} = 0.333 \times {}^0G^{fcc}_{Au} + 0.667 \times {}^0G^{hcp}_{Tb} - 65828 + 5.648T$

Table 2 Comparison between the calculated and experimental results[14-16] in the Au-Tb system

Invariant reaction	Reaction type	Composition at.% Tb			Temperature, °C	References
(βTb) ↔ (αTb) + L	Catatectic	≈97.5	> 99	≈91.5	1210	16
		98.7	99.9	94.1	1210	This work
L ↔ (αTb) + $AuTb_2$	Eutectic	80.0	> 99	66.7	880	16
		81.4	99.6	66.7	880	This work
L + αAuTb ↔ $AuTb_2$	Peritectic	71.0	50.0	66.7	1000	16
		70.0	50.0	66.7	1000.1	This work
L ↔ βAuTb	Melting		50.0		1590	16
			50.0		1623 ± 10	15
			50.0		1589.4	This work
βAuTb ↔ αAuTb	Polymorphic		50.0		1450	16
			50.0		1450.1	This work
L +αAuTb ↔ Au_4Tb_3	Peritectic	41.0	50.0	42.9	1340	16
		40.8	50.0	42.9	1340.1	This work
L + Au_4Tb_3 ↔ $Au_{10}Tb_7$	Peritectic	38.0	42.9	41.2	1210	16
		37.3	42.9	41.2	1210.1	This work
L ↔ $Au_{10}Tb_7$ + Au_2Tb	Eutectic	37.5	41.2	33.3	1195	16
		37.0	41.2	33.3	1199.6	This work
L ↔ Au_2Tb	Melting		33.3		1265	16
			33.3		1264.9	This work
L ↔ Au_2Tb + Au_3Tb	Eutectic	29.0	33.3	25	1160	16
		28.8	33.3	25	1162.2	This work
L ↔ Au_3Tb	Melting		25.0		1215	16
			25.0		1215.5	This work
L + Au_3Tb ↔ $Au_{51}Tb_{14}$	Peritectic	21	25.0	21.54	1175	16
		21.53	25.0	21.54	1172.4	This work
L + $Au_{51}Tb_{14}$ ↔ Au_6Tb	Peritectic	12.0	21.5	14.3	855	16
		12.1	21.5	14.3	856.5	This work
L ↔ Au_6Tb + (Au)	Eutectic	11.0	14.3	1.5	805	16
				1.5	809	14
		9.5	14.3	2.1	803.5	This work

(Au_6Tb, $Au_{51}Tb_{14}$, Au_3Tb, Au_2Tb, $Au_{10}Tb_7$, Au_4Tb_3, αAuTb, βAuTb and $AuTb_2$ phases). Firstly, Gschneidner et al.[13] established the original Au-Tb phase diagram, where there are six intermetallic compounds: AuTb, Au_2Tb and Au_3Tb form congruently, whereas $AuTb_2$, $Au_{51}Tb_{14}$ and Au_6Tb form peritectically. The maximum solid solubility of Tb in the fcc (Au) was found to be about 1.5 at.% by Rider et al.[14] with using the x-ray parametric method. Then McMasters et al.[15] reported the melting points of the equiatomic compound AuTb phase. Later, with the differential thermal analysis, x-ray diffraction, optical microscopy, scanning electron microscopy and electron probe microanalysis, Saccone et al.[16] constructed the phase diagram of the Au-Tb system and presented four eutectic reactions: L (80.0 at.% Tb) ↔ (αTb) + $AuTb_2$ at 880 °C, L (37.5 at.% Tb) ↔ $Au_{10}Tb_7$ + Au_2Tb at 1195 °C, L (29.0 at.% Tb) ↔ Au_2Tb + Au_3Tb at 1160 °C, and L (11.0 at.% Tb) ↔ Au_6Tb + (Au) at 805 °C and five peritectic reactions: L + αAuTb ↔ $AuTb_2$ at 1000 °C, L + αAuTb ↔ Au_4Tb_3 at 1340 °C, L + Au_4Tb_3 ↔ $Au_{10}Tb_7$ at 1210 °C, L + Au_3Tb ↔ $Au_{51}Tb_{14}$ at 1175 °C, and L + $Au_{51}Tb_{14}$ ↔ Au_6Tb at 855 °C. The three stoichiometric compounds of βAuTb, Au_2Tb and Au_3Tb melt congruently at 1590, 1265, and 1215 °C respectively.[16] According to Saccone's work,[16] the solid solubility of gold in the (αTb) phase was estimated to be less than 1.0 at.%, and the temperature of the catatectic equilibrium ((βTb) ↔ (αTb) + L) was suggested to be about 1210 °C. Based on the previous work,[13-16] Okamoto[17] critically reviewed the phase diagram of Au-Tb binary system.

In addition, the enthalpies of formation of the compounds αAuTb, Au_2Tb, and Au_3Tb phases were carried out by Meschel and Kleppa[18] with the high-temperature direct synthesis calorimetry. De Boer and Pettifor[8] predicted that the value of the enthalpy of mixing of liquid phase with the reference states of pure liquid metals at 50 at.% Tb in the Au-Tb system is −74 kJ/mol, however, no temperature dependence was given in the work of de Boer and Pettifor.[8]

3.1.2 Thermodynamic Assessments. Figure 1 shows the calculated phase diagram of the Au-Tb system: (a) from Ref 6; (b) in this work compared with the experimental data.[14-16] Compared to the preliminary result,[6] the present calculated results are in more reasonably agreement with the experimental data. The calculated composition of liquid phase in the catatectic equilibrium ((βTb) ↔ (αTb) + Liq) is 94.1 at.% Tb, which is slightly larger than the literature

data (≈91.5 at.% Tb).[16] Because the composition of the Tb-rich catatectic reaction have not been confirmed by precisely experiments,[16] which need further experiments in the future.

A set of complete self-consistent thermodynamic parameters describing the Gibbs free energy of each phase in the Au-Tb system is given in Table 1. The calculated compositions and temperatures for the invariant reactions are compared with the experimental data in Table 2, which shows a good agreement.

The calculated standard enthalpies of formation of the compounds at 298 K with reference states of the Au (fcc) and Tb (hcp) phases compared with the experimental data[18] are presented in Fig. 2. The value of the standard enthalpies of formation of the αAuTb phase is slightly more positive than the experimental one.[18] Furthermore, Fig. 3 displays the calculated enthalpies of mixing of the liquid phase at 2273 K with reference states of pure Au (liquid) and Tb (liquid) in the Au-Tb binary system. The calculated value at 50 at.% Tb is −72.6 kJ/mol, which is close to the predicted values of −74 kJ/mol.[8] The obtained thermodynamic properties are also in agreement with the measured data.[8,18] The calculated entropies of mixing of liquid phase at 2273 K with reference states of pure Au (liquid) and Tb (liquid) in the Au-Tb binary system are illustrated in Fig. 4. It is seen that the entropies of mixing of liquid phase are positive in the whole composition range.

3.2 The Au-Lu System

3.2.1 Literature Information. Many researchers[14,15,19,20] investigated the Au-Lu binary system, which is consisted of three solution phases ((liquid, fcc (Au) and hcp (Lu) phases) and five intermetallic compounds (Au_4Lu, Au_3Lu, Au_2Lu, AuLu and $AuLu_2$ phases). Different from several other rare earth-gold compounds, the AuLu phase just exists in one form and melts at 1780 °C.[15] With x-ray parametric method, Rider et al.[14] reported that 7.7 at.% Lu is soluble in (Au) at the eutectic temperature of 890 °C. Subsequently, Moffatt[19] had constructed a hypothetical phase diagram with almost no knowledge of the liquidus

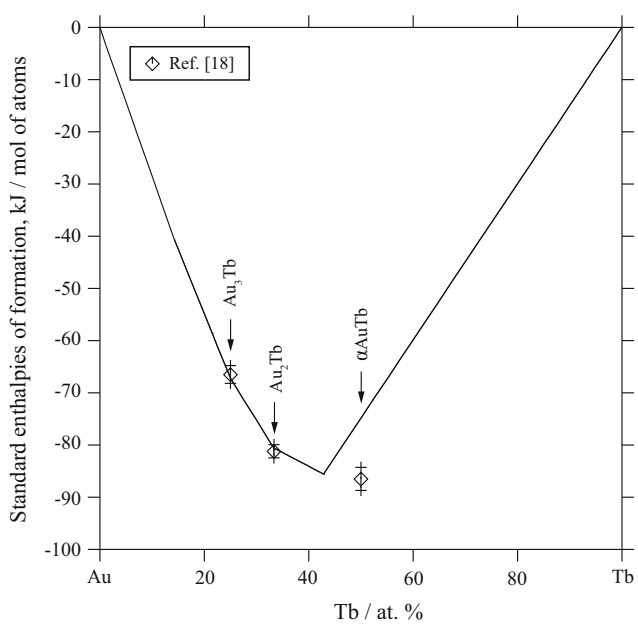

Fig. 2 Calculated standard enthalpies of formation at 298 K with reference states of Au (fcc) and Tb (hcp) compared with the experimental data[18]

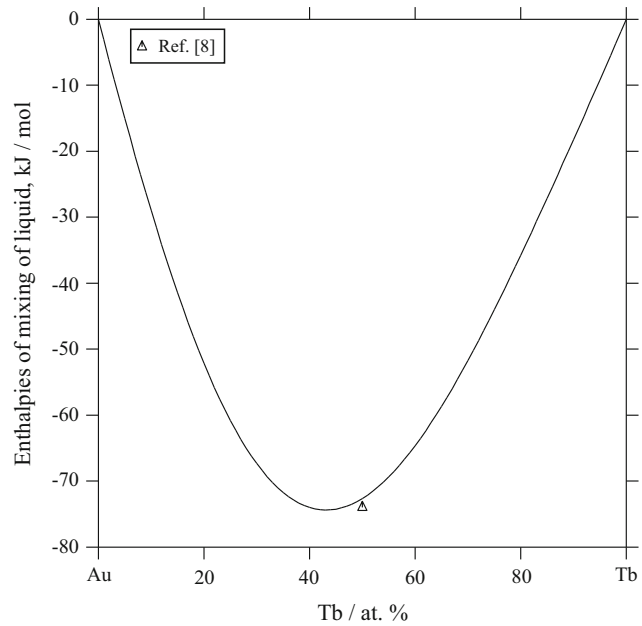

Fig. 3 Calculated enthalpies of mixing of liquid phase at 2273 K with reference states of pure elements of Au (liquid) and Tb (liquid) compared with the predicted data[8]

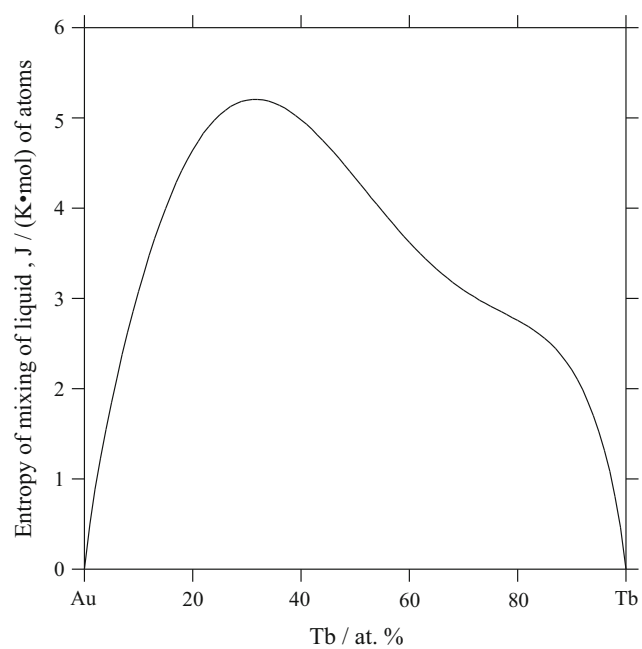

Fig. 4 Calculated entropies of mixing of liquid phase at 2273 K with reference states of pure elements of Au (liquid) and Tb (liquid)

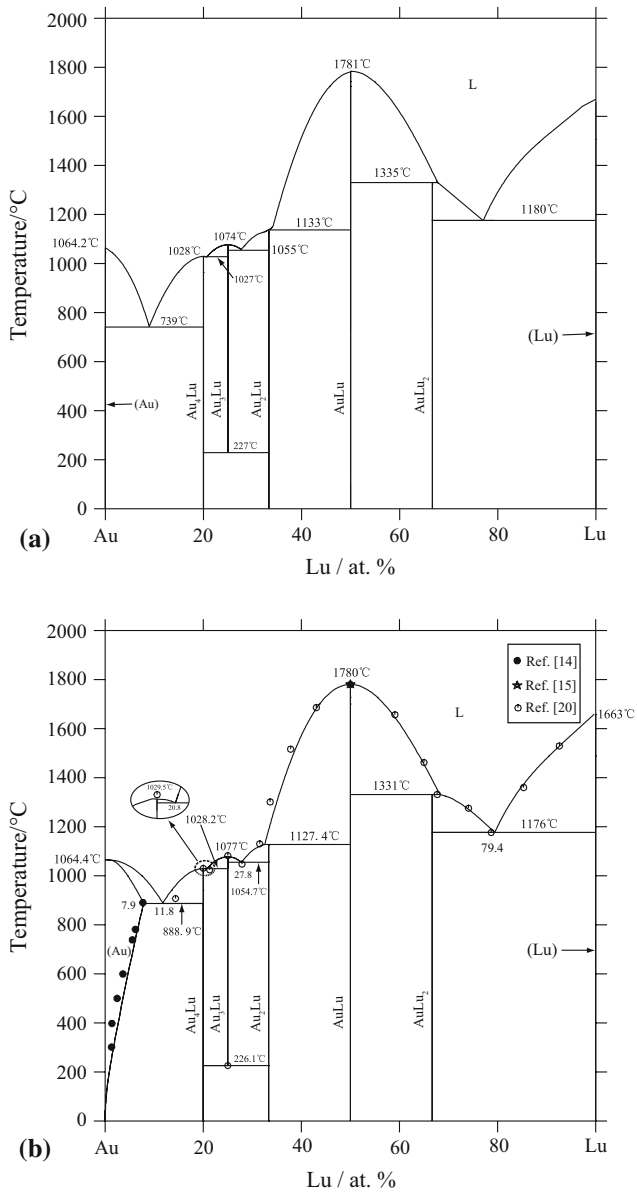

Fig. 5 Calculated phase diagram of the Au-Lu system: (a) from Ref 7; (b) in this work compared with the experimental data[14,15,20]

temperatures. According to the previous work, Gschneidner et al.[20] established the entire Au-Lu phase diagram, where there are four eutectic reactions: L (11.4 at.% Lu) ↔ Au_4Lu + (Au) at 742 °C, L (21.3 at.% Lu) ↔ Au_3Lu + Au_4Lu at 1024 °C, L (28.0 at.% Lu) ↔ Au_3Lu + Au_2Lu at 1047 °C and L (78.7 at.% Lu) ↔ $AuLu_2$ + (Lu) at 1176 °C, and two peritectic reactions: L (31.5 at.% Lu) + AuLu ↔ Au_2Lu at 1130 °C and L (67.7 at.% Lu) + AuLu ↔ $AuLu_2$ at 1331 °C.

Additionally, Fitzner and Kleppa[21] measured the standard enthalpies of formation of the Au_4Lu, Au_3Lu and AuLu phases by using a high-temperature microcalorimeter. With the same method, Meschel and Kleppa[18] measured the standard enthalpy of formation of Au_2Lu phase. De Boer and Pettifor[8] predicted that the value of the enthalpy of mixing of liquid phase with the reference states of pure liquid metals at 50 at.% Lu in the Au-Lu system is −75 kJ/mol, however, no temperature dependence was given in the work of de Boer and Pettifor.[8]

Table 3 Thermodynamic parameters in the Au-Lu binary system assessed in the present work

Parameters in each phase, J/mol
Liquid phase, format (Au, Lu)
${}^0L^{Liq}_{Au,Lu} = -292{,}100 - 3T$
${}^1L^{Liq}_{Au,Lu} = -86{,}670$
${}^2L^{Liq}_{Au,Lu} = 59{,}930$
${}^3L^{Liq}_{Au,Lu} = 7000 + 19.3T$
Fcc phase, format (Au, Lu)
${}^0G^{fcc}_{Lu} = {}^0G^{hcp}_{Lu} + 5000$, Ref 22
${}^0L^{fcc}_{Au,Lu} = -278{,}000 + 2.8T$
${}^1L^{fcc}_{Au,Lu} = -32{,}300 - 14T$
${}^2L^{fcc}_{Au,Lu} = 32{,}900$
$AuLu_2$ phase, format $(Au)_{0.333}(Lu)_{0.667}$
${}^0G^{AuLu_2}_{Au:Lu} = 0.333 \times {}^0G^{fcc}_{Au} + 0.667 \times {}^0G^{hcp}_{Lu} - 67788 + 1.31T$
AuLu phase, format $(Au)_{0.5}(Lu)_{0.5}$
${}^0G^{AuLu}_{Au:Lu} = 0.5 \times {}^0G^{fcc}_{Au} + 0.5 \times {}^0G^{hcp}_{Lu} - 99020 + 0.5T + 0.49TLn(T)$
Au_2Lu phase, format $(Au)_{0.667}(Lu)_{0.333}$
${}^0G^{Au_2Lu}_{Au:Lu} = 0.667 \times {}^0G^{fcc}_{Au} + 0.333 \times {}^0G^{hcp}_{Lu} - 82086 + 1.1T + 0.48TLn(T) - 0.53 \times 10^{-3}T^2$
Au_3Lu phase, format $(Au)_{0.75}(Lu)_{0.25}$
${}^0G^{Au_3Lu}_{Au:Lu} = 0.75 \times {}^0G^{fcc}_{Au} + 0.25 \times {}^0G^{hcp}_{Lu} - 67420 + 1.1T + 0.1TLn(T) - 0.11 \times 10^{-3}T^2$
Au_4Lu phase, format $(Au)_{0.8}(Lu)_{0.2}$
${}^0G^{Au_4Lu}_{Au:Lu} = 0.8 \times {}^0G^{fcc}_{Au} + 0.2 \times {}^0G^{hcp}_{Lu} - 60225 + 1.67T + 0.31TLn(T)$
${}^0G^{fcc}_{Lu}$ is taken from the data given in Ref 22

3.2.2 Thermodynamic Assessments. Figure 5 illustrates the calculated phase diagram of the Au-Lu system: (a) from Ref 7; (b) in this work compared with the experimental data.[14,15,20] It is apparent that the present results agree better with the experimental data[14,15,20] than the preliminary result.[7] The calculated temperature of the invariant eutectic reaction (L ↔ Au_3Lu + Au_2Lu) is 7.7 °C higher than the experimental data.[20] Since up to now there is insufficient experimental information in this region, which needs further accurate investigation in the future, thus the calculated result is considered acceptable to an extent. Thermodynamic parameters of all phases for the Au-Lu system are presented in Tables 3 and 4 lists all the assessed and experimental invariant equilibria in the Au-Lu system. The assessed maximum terminal solubility of Lu in (Au) is 7.9 at.% Lu, which agrees with the experimental data[14] mentioned above.

Table 4 Comparison between the calculated and literature results[14,15,20] in the Au-Lu system

Invariant reaction	Reaction type	Composition at.% Lu			Temperature, °C	References
$L \leftrightarrow (Lu) + AuLu_2$	Eutectic	78.7	100.0	66.7	1176	20
		79.4	100.0	66.7	1176	This work
$L + AuLu \leftrightarrow AuLu_2$	Peritectic	67.7	50.0	66.7	1331	20
		68.2	50.0	66.7	1331	This work
$L \leftrightarrow AuLu$	Melting		50.0		1780	20
			50.0		1780 ± 10	15
			50.0		1779.6	This work
$L + AuLu \leftrightarrow Au_2Lu$	Peritectic	31.5	50.0	33.3	1130	20
		32.5	50.0	33.3	1127.4	This work
$L \leftrightarrow Au_2Lu + Au_3Lu$	Eutectic	27.9	33.3	25.0	1047	20
		27.8	33.3	25.0	1054.7	This work
$L \leftrightarrow Au_3Lu$	Melting		25.0		1082	20
			25.0		1077	This work
$L \leftrightarrow Au_3Lu + Au_4Lu$	Eutectic	21.3	25	20.0	1024	20
		20.8	25	20.0	1028.2	This work
$L \leftrightarrow Au_4Lu$	Melting		20.0		1030	20
			20.0		1029.5	This work
$Au_3Lu \leftrightarrow Au_2Lu + Au_4Lu$	Eutectoid	25.0	33.3	20.0	226	20
		25.0	33.3	20.0	226.1	This work
$L \leftrightarrow Au_4Lu + (Au)$	Eutectic	11.4	20.0	0	742	20
				7.7	890	14
		11.8	20.0	7.9	888.9	This work

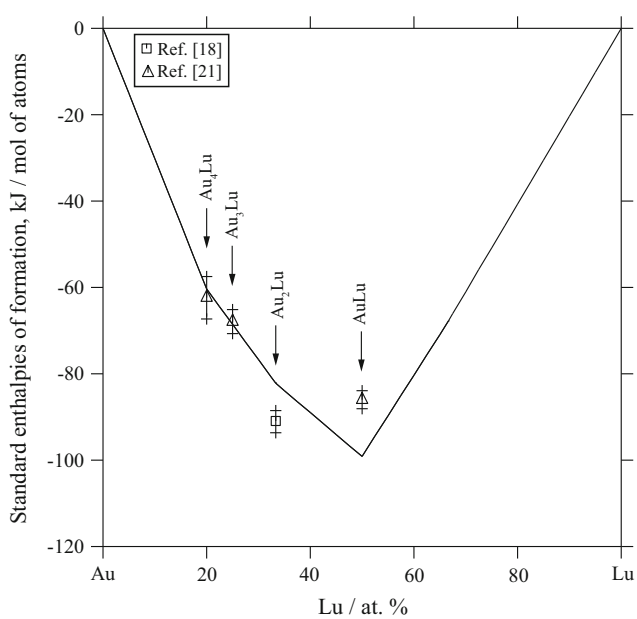

Fig. 6 Calculated standard enthalpies of formation at 298 K with reference states of Au (fcc) and Lu (hcp) compared with the experimental data[18,21]

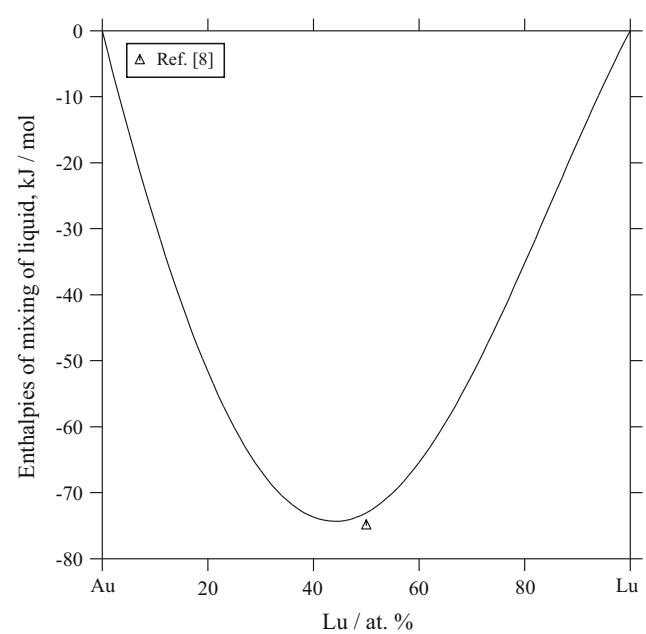

Fig. 7 Calculated enthalpies of mixing of liquid phase at 2273 K with reference states of pure elements of Au (liquid) and Lu (liquid) compared with the predicted data[8]

Figure 6 plots the calculated standard enthalpies of formation at 298 K with reference states of the Au (fcc) and Lu (hcp) phases in comparison with the experimental data.[18,21] The calculated standard enthalpies of formation of Au_2Lu and AuLu phases seem not identical with the experimental data, where the calculated result of AuLu phase is slightly more negative[21] while the Au_2Lu phase is less negative.[18] By considering the agreements of the most of the experimental data including the data of phase

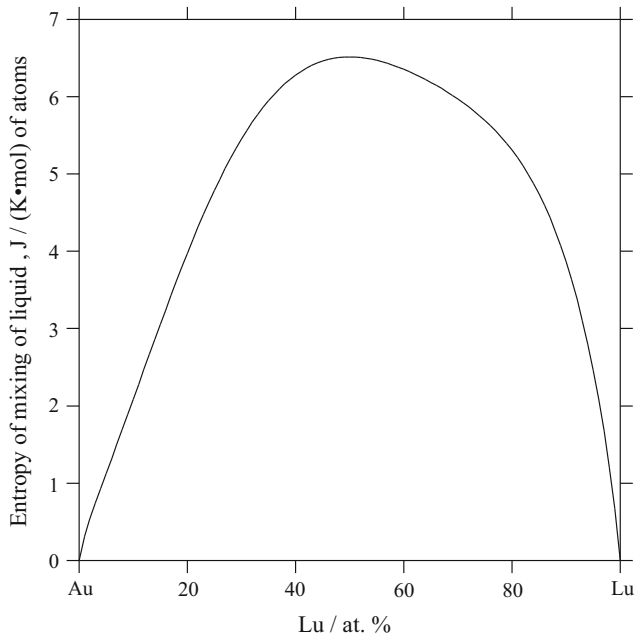

Fig. 8 Calculated entropies of mixing of liquid phase at 2273 K with reference states of pure elements of Au (liquid) and Lu (liquid)

equilibria and thermodynamic properties, the present calculation is still acceptable. Figure 7 shows the calculated enthalpies of mixing of the liquid phase at 2273 K with reference states of pure Au (liquid) and Lu (liquid) in the Au-Lu binary system. The calculated value at 50 at.% Lu is −73.2 kJ/mol, which is reasonably close to the predicted data.[8] Therefore, the calculated enthalpies of mixing of the liquid phase in the Au-Lu system are reasonable. Figure 8 presents the calculated entropies of mixing of liquid phase at 2273 K with reference states of pure Au (liquid) and Lu (liquid) in the Au-Lu binary system. It is seen that the entropies of mixing of liquid phase are positive in the whole composition range.

4. Conclusions

The phase diagrams in the Au-Tb and Au-Lu binary systems have been thermodynamically optimized by combining with the experimental information including phase equilibria and thermodynamic properties. A coherent set of thermodynamic parameters has been obtained and most of the experimental information can be satisfactorily reproduced based on the optimized thermodynamic parameters.

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